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Stochastic behaviour of Ghana power grid with massive electro-mobility penetration

Emmanuel Yeboah Asuamah [a,b,*], Abass Ishaq [a,c], Samuel Jonas Yeboah [a], William Duodu Asihene [d]

[a] School of Engineering, Department of Electrical and Electronic Engineering, Sunyani Technical University, P. O. Box 206, Sunyani, Bono Region, Ghana
[b] Regional Centre for Energy and Environmental Sustainability, School of Energy, University of Energy and Natural Resources (UENR), P. O. Box 214, Sunyani, Ghana
[c] Northern Electricity Distribution Company (NEDCo), Ghana
[d] Cenpower Company Limited, Tema, Ghana

A R T I C L E I N F O

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A B S T R A C T

The study focused on determining how Ghana Grid will behave stochastically with massive Electro-Mobility Penetration. The study employed Microsoft Excel and DIgSILENT PowerFactory tools to forecast the EVs, electricity demand, and grid impact analysis. A well-to-wheel approach was adopted to analyse the emission reduction potential of EVs. A key issue was whether Ghana's current electricity generation capacity would be able to sustain the extra energy demand that the addition of EVs comes with. Analysis of data in this study found that Ghana's current electricity generation will be able to handle the increase in demand before 2028, allowing for grid planning. The result shows that the grid lines were able to accommodate the EV growth however the transformer could not accommodate it. It is important to note that the attributes of electrical distribution networks are unique to the system under consideration. As a result, it can be misleading to relate the findings from two cases to all low-voltage distribution grids. However, the results of these scenarios can be used to show low-voltage grids with similar characteristics. The emission analysis presented good figures depicting that EVs emit only 62.803 gCO_2/km, signifying a significant reduction in greenhouse gas emissions compared to their counterpart. This is the case because of the generation mix of Ghana (34.1% from hydro against 65.3% from thermal and 0.55% from renewables). The results underscored the imperative need for proactive grid planning and augmentation to accommodate the anticipated surge in EV adoption. Furthermore, the study illuminated a pressing concern of demand exceeding supply before 2028. This research thus provides an essential roadmap for informed decision-making and grid

Abbreviations: EVs, electric vehicles; BEVs, battery electric vehicles; PHEV, plug-in hybrid electric vehicles; HEV, hybrid electric vehicles; CO_2, carbon dioxide; CH4, methane; CO, carbon monoxide; N2O, nitrous oxide; ICE, internal combustion engine; GHGs, Greenhouse Gases; SDGs, sustainable development goals; KWh, kilo-watts hour; GWh, giga watt hour; TWh, terabyte watt hour; Ktoe, kilo tonnes of oil equivalent; V2G, vehicle to grid; G2V, grid to vehicle; MW, mega watt; VRA, volta river authority; GHS, Ghana cedis; PURC, public utilities regulatory commission; EPA, environmental protection agency; RE, renewable energy.

* Corresponding author at: Regional Centre for Energy and Environmental Sustainability, School of Energy, University of Energy and Natural Resources (UENR), P. O. Box 214 and School of Engineering, Department of Electrical and Electronic Engineering, Sunyani Technical University, P. O. Box 206, Bono Region, Sunyani, Ghana.

E-mail address: emmanuel.yeboah@uenr.edu.gh (E.Y. Asuamah).

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management strategies to support sustainable EV adoption while maintaining grid stability and reliability.

Introduction

The global energy landscape has encountered a significant challenge in meeting the growing power demands while ensuring sustainable energy provision. This challenge primarily arises from the finite availability of fossil fuels such as oil, natural gas, and coal. The rapid expansion of technology and industrial sectors, coupled with the subsequent surge in energy consumption, has intensified reliance on fossil fuels, resulting in escalating energy costs [1–3]. Moreover, concerns about environmental issues, including rising global temperatures, greenhouse gas (GHG) emissions, and the depletion of non-renewable resources, have become prominent global concerns [2]. Each day, substantial amounts of carbon dioxide are released into the atmosphere, with industrial sectors and transportation fleets accounting for the majority of CO_2 emissions, based on the International Energy Agency report [4].

Alongside electricity generation, the transport sector is the largest source of greenhouse gas emissions in the world [5]. Incomplete burning of fossil fuels by vehicles causes climate change that is increasingly posing severe risks to ecosystems and human health such as the rise in sea levels, global warming, wildfires, flooding, extreme droughts, and tropical storms.

With the evident signs of climate change, governments and organizations worldwide are increasingly recognizing the potential of electrifying the transportation sector as a means to mitigate greenhouse gas emissions. This shift entails a transition from traditional oil-based transportation to vehicles powered by electric grids, off-grid systems, or hydrogen-based solutions [6]. The decarbonization of transportation fleets has become a pivotal focus in the energy sector, particularly in the most developed nations [7–9]. As a potential solution, there is a growing inclination towards Integrating renewable energy sources (RESs) with electric vehicles (EVs) and vehicle-to-grid (V2G) systems within smart grids. This integration aims to minimize dependence on petroleum products, meet power demands, and subsequently mitigate concerns related to global warming [10–12]. This has led the scientific community to research and develop an alternative source of vehicle power to reduce environmental harm.

Hybrid and electric vehicles have gained popularity over the past few years. Numerous studies proved that hybrid and electric vehicles provide a clean solution compared to fossil-fuel-powered vehicles. EVs are expected to significantly reduce road transport emissions, given an increasingly renewable power generation [13]. Electric transportation offers ideal opportunities for the broader introduction of renewables to the transport sector [14]. Apart from mitigating climate change and air pollution by reducing emissions, electric vehicles provide substantial efficiency improvements and long-term cost advantages. There is a noticeable decline in the demand for fossil fuel vehicles, as consumers are becoming increasingly conscious of these environmental concerns. Given the significant surge in fuel prices over recent decades, it is evident that Electric Vehicles (EVs) are the future of transportation [15].

The increasing concentration of greenhouse gases (GHGs) in the atmosphere, primarily caused by emissions, is a pressing global concern. The transportation sector is a significant contributor to GHG emissions, with CO_2 emissions exhibiting a consistent upward trend, resulting in a critical worldwide issue [16]. In Ghana, the continuous growth in vehicle population and fuel consumption has led to a corresponding increase in GHG emissions, from 1.38 $MtCO_2e$ in 1990 to 19.40 $MtCO_2e$ in 2020 [17]. If no significant changes in policies or economic factors occur, this trend is projected to further escalate by 2040, as the existing patterns for meeting road transport and energy demands in Ghana continue [18]. Many countries and individuals are increasingly recognizing the environmental implications of these emissions and are taking steps to invest in alternative energy sources for vehicles. The adoption of electric vehicles (EVs) has been on the rise, including in Ghana, where several EV companies have emerged to cater to the growing demand. However, the increased adoption of EVs in Ghana will have implications for the country's annual electricity demand, which poses challenges given the current state of energy production and intermittent power outages. The uptake of EVs in Ghana requires the development of supporting infrastructure for a smooth transition, which is presently lacking.

Currently, EVs in Ghana rely on the national power grid for charging, where renewable energy only accounts for about less than 1% of the supply and more than half is derived from thermal power plants that, over time, add to the greenhouse effect. This indicates that using an EV in Ghana does not significantly contribute to cutting down on emissions. In line with the sustainable development goals, electro-mobility is related to the following specific SDGs; Goal 13: Climate Action, Goal 3: Good Health and Well-Being, Goal 7: Affordable and Clean Energy, Goal 9: Industry, Innovation and Infrastructure, Goal 11: Sustainable Cities and Communities and Goal 17: Partnerships for the Goal. This study investigates the stochastic behaviour of Ghana Grid with Massive Electro Mobility Penetration and their emission reduction potential.

Electric vehicle adoption in Ghana

The International Trade Centre reveals that between the start of 2017 and the end of 2021, Ghana received approximately 17,660 Plug-in Electric Vehicles (PEVs). During the same period, the country imported a total of 9431 motorized Electric two & three-wheelers (E2&3W) units, the vast majority (98%) of which were Chinese-sourced Battery Electric Vehicles (BEVs) [19]. Ghana Revenue Authority (GRA) in its 2021 report stated that standard hybrid-electric vehicles (HEVs) account for 91.5% of all Plug-in Electric Vehicles in the country, trailed by BEVs and plug-in hybrid electric (PHEVs) at 5.1% and 3.3%, respectively [20].

Meanwhile, Ghana's automobile industry has been identified as one of the fastest-growing globally, contributing a quarter to the nation's GDP. Despite the global pandemic in 2020, 4710 new passenger vehicles were registered in Ghana, marking a 17% decrease from the previous year's sales [21]. The number of cars in Ghana and across the globe is anticipated to double by 2040. Consequently, if all vehicles in Ghana are replaced with EVs, the country's electricity supply, which currently stands at 4132 Megawatts from mixed sources, will need to double to meet this demand [22].

Electric vehicle and power sector challenges

Integrating electric vehicles (EVs) into the power sector presents opportunities and challenges, particularly in developing countries. Key focus areas include tariff design, charging infrastructure, renewable energy source (RES) coupling, and the complementarities between these elements. Effective tariff structures are essential for managing EV charging demand and promoting grid stability. Minimising expensive grid upgrades and providing effective management of EV charging, submetering offering customers access to EV-specific rates without requiring the installation of a separate meter emerges as an attractive solution for both EV owners and grid operators [23]. The same study discusses innovative rate designs that encourage flexible EV charging, aligning consumption with periods of low demand or high renewable generation as an effective approach to managing the grid [23]. Similarly, [24] explores network tariff designs that integrate distributed energy resources and EVs, highlighting the importance of dynamic pricing in optimising grid operations.

The development of adequate charging infrastructure is critical for EV adoption. Deployment of such infrastructure should be guided by a strategy that accounts for the surrounding environment and the behaviour patterns of electric vehicle users. Neglecting these factors could result in unmet user needs and increased operational costs [25]. This calls for strategic placement and capacity planning to meet the future demand of users and reduce operational costs. In the Canary Islands, a simulation of Tenerife's 2040 traffic scenario was conducted using MATSim (Multi-Agent Transport Simulation) to identify optimal locations and the required number of charging points for the network. This simulation aimed to evaluate Tenerife's electromobility infrastructure plan and offer insights into effective strategies for deploying charging infrastructure [26].

Coupling EVs with renewable energy sources can enhance sustainability and reduce emissions, especially in developing countries where rate of energy access is generally low. The integration of rooftop solar photovoltaic (PV) systems and electric vehicles (EVs) in urban areas is increasingly recognized as a cost-effective strategy for simultaneously decarbonizing the energy, transport, and building sectors. Research highlights the potential of this approach to reduce greenhouse gas emissions while offering significant economic benefits. As a result, policymakers are encouraged to develop and implement integrated rooftop solar PV and EV decarbonization policies. Such measures could capitalize on the synergies between these technologies, enhancing sustainability and supporting broader urban climate goals [27]. Electric vehicles (EVs) can interact with the grid in two ways: charging and discharging. PV -batteries are advanced technology where batteries store electricity surplus produced by PVs during the day, avoiding curtailment and restore it to the grid to shave peak load or when external grid constraints are identified. Coupling PV-EV as seen in emerging technologies such as vehicle-to-grid and blockchain can use the EV as a battery source and return stored power to the grid when needed and thus forces electricity grid operators to readjust the way how the grid is managed [23,28]. Thompson and Perez (2020) explore Vehicle-to-Anything (V2X) energy services, identifying value streams and regulatory implications of using EVs as distributed energy resources [29]. Venegas et al. (2021) examine barriers to the participation of distributed energy resources, including EV fleets, in local flexibility markets, highlighting the need for supportive frameworks to enable active grid integration [30].

As per a study conducted in California, it is projected that by 2040, if the entire fleet of 31 million cars in the state is converted to electric vehicles, there will be 22,281,720 plug-in hybrid electric vehicles (PHEVs) and 8718,720 pure electric vehicles (PEVs) on the road. Of all PEVs, 31% will require two battery charges per day, the same percentage is estimated for PHEVs. The average battery capacity of a typical PEV is 24 KWh and for a PHEV it's 8.5 KWh, meaning an average PEV needs 24 KWh for a full charge and a PHEV needs 8.5 KWh [15]. Given these figures, a total of 797.26668 GWh of electricity per day is needed to power California's entire EV fleet in 2040. As a result, California will need to supply 291 TWh of electricity in 2040, which is 80% of its projected electricity consumption, to power its fleet of only PEVs and PHEVs.

Impact of electric vehicles on the SDGs

Energy use in transport has increased significantly in correlation with world population growth and economic development. The production and use of combustion engine vehicles contribute to climate change negatively. The transportation sector accounts for 14% of global greenhouse gas (GHG) emissions [31]. This infringes upon the Sustainable Development Goals (SDGs) which advocate for a greener environment and environmentally friendly source of energy devoid of poisonous emissions. The mass investment and usage of Electric Vehicles (EVs) in Ghana are very vital for the achievement of the United Nations Global Sustainable Development Goals (SDGs) due to the increasing demand for mobility and freight transport, without successful green technology and green behaviour transitions, the energy intensity and environmental impact of transport may increase substantially for the worse [32]. EV is fuelled by energy and is therefore directly linked to:

1. SDG #7: Affordable and Clean Energy
2. SDG#13: Climate Action
3. SDG#9 Industry, Innovation, and Infrastructure

This initiative is expected to support the actualization of virtually all the goals, including those associated with economic and social development, connecting goods with markets, supporting agricultural productivity, and access to services.

Ghana's energy supply and demand systems situations

Ghana's principal energy sources encompass electricity, biomass, and fossil fuels, with local energy generation majorly relying on sources such as biomass, hydroelectric power stations, thermal power facilities, and solar power [33]. To satisfy Ghana's energy needs, there is an importation of electricity, fossil fuels, and crude oil, supplementing the main domestic energy production. This imported energy fuels diverse sectors in Ghana, including residential, commercial services, agriculture and fisheries, transportation, and industry [34].

There has not been much change in the current energy supply and framework with the reliance on legacy energy supply systems as the means of generation and production of energy (Eshun & Amoako-tuffour, 2016). The energy supply and demand systems in Ghana have gone through many phases: starting with diesel generators and stand-alone electricity supply systems owned by industrial mines and factories, to the hydro phase following the construction of the Akosombo dam, and now supplementary supply from the thermal complement phase powered by gas and/or light crude oil [35]. But, according to Somoah (2018), there has been the severity of the energy supply that threatens the country's economic growth and transformation. Thus, their current power generation systems have proved not self-sufficient and reliable both for industrial and domestic usage.

The current energy installed generation capacity available for grid power supply at the transmission level as of the end of 2018 was approximately 4980 Megawatt (MW). The composition of the total primary embedded generation including hydro, thermal, solar plants, and diesel generation power at the sub-transmission level is included.

Materials and method

The approach utilised to assess Ghana's preparedness to transition all traditional vehicles into electric vehicles for transportation purposes was based on software simulations.

Specifically, current data for Ghana was collected, theses data include the number of imported and registered EVs, Ghana's population, the nation's electricity generation, and consumption, and the number, and type of generation stations from relevant stakeholders such as the Driver Vehicle Licensing Authority (DVLA), and EV companies. Again, data was collected on transportation from the Ministry of Transport, Ghana Energy Commission, Ghana Revenue Authority, and Ghana Statistical Urgency. The following data were taken, current EV technologies, EV policies, renewable energy generation, and greenhouse emissions. Useful articles and reviews from various facets were considered.

Based on the historical trends of these data, data was forecasted and analysed. The first was to analyse how energy demand for EV charging contributes to peak demand. This assessment of the national energy demand was done using Microsoft Excel Tool. The study then modified the number of EVs and their respective charging techniques to evaluate their effect on greenhouse gas emissions, and grid stability. The assessment exposed how different EV charging strategies impact the local low-voltage grids. This involved carrying out case studies on a feeder and simulating the impact of EV charging through power flow analysis in PowerFactory.

Data analysis

Data analysis is an integral process that encompasses the gathering, cleansing, transformation, and modelling of data to extract valuable insights, formulate conclusions, and aid decision-making. In the frame of the research, two potent tools were employed: Microsoft Excel and DIgSILENT PowerFactory 15.1.7 $\times$ 86.

Microsoft Excel is a popularly used spreadsheet software recognized for its powerful capabilities in data handling, computations, and graphing tools. It provides an intuitive interface, enabling researchers to execute tasks from elementary data logging to intricate statistical analysis. Microsoft Excel is a widely utilized tool for forecasting across various domains due to its accessibility and user-friendly interface. Its built-in functions and add-ins facilitate a range of forecasting methods, from simple linear projections to more complex time-series analyses. The Analysis Toolpak add-in in Microsoft Excel was utilized to forecast demands using three models: Auto-Regressive, Moving Average, and Exponential Smoothing. The accuracy of these forecasts was subsequently evaluated using three prediction metrics: Mean Absolute Deviation (MAD), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE) to determine which model among three presents the best result [36]. In India a Short-term forecasting of total energy consumption was conducted using 1997–2003 Worksheet with GDP, population and GDP per capita as input values or factors. According to the author, Microsoft Excel was chosen because it utilizes basic yet reliable methodologies that, for certain simple research problems such as forecasting energy consumption in the Republic of India can outperform advanced methods, which are often built upon these foundational approaches [37]. Microsoft Excel is widely utilized due to its user-friendly interface. It is particularly favored for tasks that require consistency checks and for processes that are challenging to automate, as it facilitates qualitative research and analysis of industry practices [38].

DIgSILENT PowerFactory 15.1.7 $\times$ 86 is a comprehensive power system analysis software utilized for a variety of projects in generation, transmission, distribution, and industrial systems. It provides a completely adjustable and programmable environment, paired with sophisticated algorithms for network analysis. PowerFactory permits users to conduct in-depth modelling and simulation of power systems, assisting researchers in comprehending complex interactions within these systems. As the purpose of the analysis is to figure out how EV uptakes will impact the national energy system, a tool that can simulate the energy system, as opposed to optimize it, is adequate for this analysis.

Table 1-1
EV vehicle growth in Ghana up to 2030 (Source: Authors' forecasted data from [40].

Year (n)	0	1	2	3	4	5	6	7	8
Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Conventional Car Population(K)	3000	3130	3260	3390	3520	3650	3780	3910	4040
EV Yearly growth	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000	30,000
EV Number In Ghana	30,000	60,000	90,000	120,000	150,000	180,000	210,000	240,000	270,000

Vehicle population statistics

The current growth in terms of the number of vehicles that are in Ghana or registered every year is used as historical growth data to estimate the number of vehicles that will be available in 2030. Historical and current trends data of the population and registered vehicles in Ghana were collected and projected using the Excel forecast function. It is anticipated that the current number of vehicles will influence the EV update. Due to this, the number of conventional vehicles in Ghana must be determined. It is expected that vehicle owners will replace their old vehicles with EVs, as a result displacing the number of different vehicle types. The Ministry of Transport (MoT) and Driver Vehicle and Licensing Authority (DVLA) do not have year-by-year vehicle population for both electric vehicles and internal combustion engine vehicles (ICE). This was a challenge for the Ghana Energy Commission in the preparation of the Ghana Electric Vehicles Baseline Survey Report 2022. However, according to data from DVLA as reported by the Energy Commission Ghana, the vehicle population in 2022 stood at 3 million and stated annual vehicle registration of 130,000 [39]. Based on the available data the vehicle population for the year 2030 was calculated using Eq. (3.1).

$$\boldsymbol{V_{Tpn}} = \boldsymbol{B_{vp}}_{2022} + ((\boldsymbol{Y_b} - \boldsymbol{Y_n}) * \boldsymbol{R_n}) \tag{3.1}$$

where

B_{vp} the total vehicle population in the nth year
B_{vp2022} the total vehicle population in 2022
Y_b the base year_2022
Y_n the nth year
R_n the annual registered vehicles

Percentage of EVs in the total registered vehicles in Ghana

Various assumptions were made in this section to determine the number of electric vehicles that would be available in Ghana in 2030 when a certain percentage of the existing automobiles are transformed into EVs. The Ghana Energy Commission has estimated 500,000 EVs in Ghana by the year 2030 [39]. However, the uptake of EVs in countries around the world has been hinged on one or more Government incentives. In other words, users of EVs are incentivized to purchase one because of the cost involved. Ghana currently does not have any incentives for the electric mobility uptake though the Energy Commission is leading a discussion with relevant institutions to consider either monetary or non-monetary incentives to promote Electric mobility updates. As a result of the shortcomings related to uptake stated above and how Ghana is slow in adopting change, it has been assumed that 1% of the total vehicle population will be replaced with electric vehicles starting from the base year 2022. The number of electric vehicles that will be available in 2030 is calculated using Eq. (3.2) and shown in Table 1-1.

$$\mathbf{EV_{TPn}} = \boldsymbol{P_{byv}}_{2022} * \boldsymbol{R_{byv}} * \boldsymbol{n} \tag{3.2}$$

where

EV_{TPn} = the total electric vehicle population in the nth year
$P_{byv2022}$ = the total vehicle population in 2022
R_{byv} = the percentage of the total vehicle population in 2022 that are electric vehicles n is the nth year

Energy consumption of the projected electric vehicles

In this study, the types of electric vehicles are classified into three to aid in the proper estimation of their energy consumption. The energy consumption (electricity to charge) of the various types is different. Electric vehicles have three major classifications namely:

1. Hybrid-electric vehicles (HEV)
2. Battery electric vehicles (BEV)
3. Plug-in hybrid electric vehicles (PHEV)

According to the survey conducted by the Ghana Energy Commission on the user's or potential users' choice of electric vehicle type, the following result was recorded.

The consumption of electric vehicles in 2030 was determined by first finding the share of each vehicle type and then multiplying it

Table 1-2
Share of electric vehicle type in total electric vehicles available [40].

Electric Vehicle Type	Percentage Share (%)
Battery electric vehicles (BEV)	69
Hybrid-electric vehicles (HEV)	23
Plug-in hybrid electric vehicles (PHEV)	8

Table 1-3
Energy consumption of different vehicle types [42].

	Average Consumption of Petrol Vehicles [kWh/km]	Average Consumption of Diesel Vehicles [kWh/km]	Average Consumption of EVs [kWh/km]	Average Petrol Consumption PHEVs [kWh/km]	Average Electricity Consumption of PHEVs [kWh/km]
Consumption	0.5278	0.5278	0.1667	0.21112	0.10002

EVs*** comprise of BEV and HEV

by the appropriate consumption rate shown in Table 1-3. The average km coverage per user per year is 60,000 km [41].

From Tables 1-2 and 1-3 the energy consumption of a vehicle type $Ev\left(\frac{kWh}{Y}\right)$ is determined through Eq. (3.3).

$$Ev\left(\frac{kWh}{Y}\right) = EV_{tpn} * V_{ts} * V_{tc} * V_{da} \tag{3.3}$$

Where
EV_{tpn} = the total EVs in the nth year.
V_{ts} = the share of EV type
V_{tc} = the vehicle type consumption
V_{da} = the average distance travelled per year
Table 1-4 shows the distribution of the yearly vehicle population and consumption according to the specific type of electric vehicle since each vehicle type has a different consumption level as described in Table 1-3

Emission analysis

Emission analysis in the EV sector encompasses a multifaceted evaluation of the environmental impacts associated with the entire lifecycle of these vehicles. It goes beyond the tailpipe emissions typically associated with gasoline or diesel vehicles and includes considerations of emissions arising from EV production, electricity generation, and end-of-life disposal. Understanding the complete emissions profile of EVs is essential for making informed policy decisions, designing sustainable transportation systems, and guiding consumers toward environmentally responsible choices [43].

Indirect emissions from electric vehicles

The greenhouse gas emissions created during the generation and transmission of electricity used to charge EV batteries, as well as the emissions connected with the production and disposal of EV components such as batteries, are referred to as indirect emissions from electric cars (EVs) [44]. The well-to-wheels CO_2 emissions for the EVs studied were computed by adding well-to-power-plant and power-plant-to-wheels emissions.

Well-to-wheels CO_2 emissions approach

The well-to-wheels CO_2 emissions analysis for EVs in Ghana was carried out using a systematic approach. To begin, the electrical mix in Ghana was established by taking into account the contribution of various energy sources to the electricity grid, such as fossil fuels (coal, oil, gas) and renewables (solar, wind, and hydro). The CO_2 emissions related to the production and distribution of the electricity utilized to charge the EVs were then calculated to quantify the well-to-power-plant emissions. Calculating the fuel efficiency of electric vehicles, the breakdown of total net power output, and the emissions for each type of fuel used in electricity generation were all part of this process.

The power-plant-to-wheels emissions were assessed by evaluating the emissions associated with converting electricity at the power plant into usable energy for the EVs and the subsequent operation of the vehicles. Power-plant-to-wheels emissions for the EV were calculated from the fuel efficiency and the average direct CO_2 emissions from electricity generation. The emission factor for Ghana is calculated based on the current electricity mix and is subject to change as the fuel sources for power generation evolve.

The well-to-wheels CO_2 emissions for EVs were obtained by combining the well-to-power-plant emissions and the power-plant-to-wheels emissions as shown in Eq. (3.4).

Table 1-4
Electric vehicle population distribution and energy consumption (Source: Authors' estimates).

Electric Vehicle Type	Percentage Share	Consumption								
		2022	2023	2024	2025	2026	2027	2028	2029	2030
BEV	69	20,700	41,400	62,100	82,800	103,500	124,200	144,900	165,600	186,300
HEV	23	6900	13,800	20,700	27,600	34,500	41,400	48,300	55,200	62,100
PHEV	8	2400	4800	7200	9600	12,000	14,400	16,800	19,200	21,600
EV (BEV+HEV)	92	27,600	55,200	82,800	110,400	138,000	165,600	193,200	220,800	248,400
Average consumption EV MWh	0.1667	276,055.2	552,110.4	828,165.6	1,104,220.8	1,380,276	1,656,331.2	1,932,386.4	2,208,441.6	2,484,496.8
AV Consumption PHEVS	0.10002	14,402,880	28,805,760	43,208,640	57,611,520	72,014,400	86,417,280	100,820,160	115,223,040	129,625,920
Total Yearly Consumption MWh		290,458.	580,916.16	871,374.24	1,161,832.32	1,452,290.4	1,742,748.48	2,033,206.56	2,323,664.64	2,614,122.72
Daily Consumption MWh		795.7	1591.5	2387.3	3183.1	3978.8	4774.6	5570.4	6366.2	7161.9

****Average Km/year travelled is 60,000

Table 1-5
Electricity generation for EVs in Ghana [40].

EV models	Electricity required to run EVs (kW h km-1)
Hyundai Kona EV	0.17
Tesla Model 3	0.21
REVAi	0.15

Table 1-6
Average CO_2 emissions from electricity generation [45].

Country	Average emissions (g CO_2 kWh-1)
Ghana	327.9

Table 1-7
Share of electricity generation from fossil fuels and renewables [46].

	Light Crude Oil	Natural gas	Hydro
Electricity Generation	11,434	2974	7521
Percentage	51.8	13.5	34.1

Table 1-8
Well-to-power-plant CO_2 emissions [47].

	Coal	Light Crude Oil	Natural gas	Nuclear
Range (gCO2 per kWh)	85–135	40–110	48–100	9–70
Mean (gCO2 per kWh)	110	75	74	40

$$W_tW_{EV}\left[g\ km^{-1}\right] = Eff_{EV}\left[MJ\ km^{-1}\right] \times (W_tPP_C + PP_tW_{EV})\ gMJ^{-1} \tag{3.4}$$

The total CO_2 emissions associated with the use of EVs throughout their life cycle were calculated, providing a comprehensive representation of emissions from energy production to vehicle operation. To assess the environmental impact of EVs, a comparison was made between the well-to-wheels CO_2 emissions of EVs and those of conventional internal combustion engine vehicles. This analysis facilitated an evaluation of the potential emission reduction achieved through the transition to EVs, highlighting the environmental benefits of adopting electric transportation in Ghana.

$$\boldsymbol{W_tW_{EV}}\left[\boldsymbol{g\ km^{-1}}\right] = \boldsymbol{Eff_{EV}}\left[\boldsymbol{MJ\ km^{-1}}\right] \times (\boldsymbol{W_tPP_C} + \boldsymbol{PP_tW_{EV}})\left[\boldsymbol{g\ MJ^{-1}}\right] \tag{3.4a}$$

W_tPP_C = well-to-power-plant emissions
PP_tW_{EV} = Power-plant-to-wheels CO_2 emissions for EVs
$W_tW_{EV}\left[g\ km^{-1}\right]$ = well-to-wheels CO_2 emissions for EVs

Well-to-power-plant CO_2 emissions

These are the CO_2 emissions generated during the production and distribution of the electricity used to charge the EVs. It includes the emissions associated with extracting, refining, and transporting the primary energy sources (such as fossil fuels or renewable energy) used to generate electricity in power plants. In the case of Ghana, the well-to-power-plant emissions would depend on the energy mix of the country, considering factors such as the contribution of fossil fuels (coal, oil, gas) and renewable sources (hydro, solar, wind) to the electricity generation.

Well-to-powerplant emissions for each EV in each country were calculated from fuel efficiency data (Table 1-5), the breakdown of total net electricity generation, and the well-to-power-plant emissions for each type of fuel used in electricity generation.

Power-plant-to-wheels CO_2 emissions

These are the CO_2 emissions produced during the conversion of electricity at the power plant into usable energy for the EVs and the subsequent operation of the vehicles. This component takes into account the efficiency of the power plant, transmission and distribution losses, as well as losses in the charging process. Additionally, it considers the efficiency of the electric drivetrain in the EV and the energy required to propel the vehicle.

Power-plant-to-wheels emissions for each EV in each country were calculated from the fuel efficiency for each EV (Table 1-5) and the average direct CO_2 emissions from electricity generation.

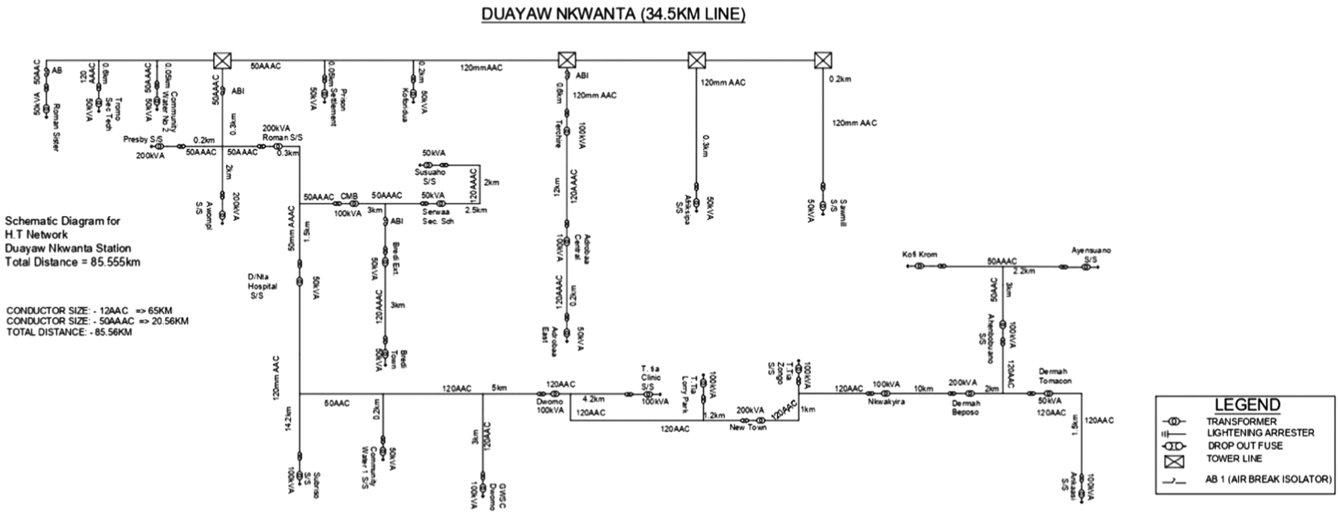

Fig. 1-1. A single line diagram of Duayaw Nkwanta distribution network (Source: NEDCo.

Table 1-9
Lines data for the distribution network (source: Duayaw Nkwanta NEDCo).

S/N	Parameter	Value	Unit
1	Rated Voltage	34.5	kV
2	Rated Current	0.4	kA
3	Nominal Cross Section	120	mmˆ2
4	Conductor Material	Aluminium	N/A
5	Phases	3	N/A
6	System Type	AC	N/A

Table 1-10
Transformer loading (Primary Side) (source: Duayaw Nkwanta NEDCo).

S/N	Parameter	Value	Unit
1.	Rated Voltage	34.5	Kv
2.	Rated Current	0.4	kA
3.	Nominal Freguency	50	Hz
4.	Cable/OHL	Overhead Line	NA
5.	System Type	AC	NA
6.	Paramter Per Length 1,2-Sequence (AC-Resistance R`(200C)	0.267	Ohm/km
7.	Reactance X`	0.203261	Ohm/km
8.	Paramter Per Length Zero Sequence (AC-Resistance R0`)	1.065	Ohm/km
9.	Reactance X0`	0.56	Ohm/km

Table 1-11
Smart meter consumption of duayaw nkwanta (source: Duayaw Nkwanta NEDCo).

Utility:	Duayaw Nkwanta	**Billing Month:**	202,301	**Printer:**	Samuel Sarquah	**Print Time:**	8/3/2023 9:47
Area		**Station**		**Customer Type**	**Tariff Bands**	**No. of Customers**	**Total Consumption kWh**
Sunyani		Duayaw Nkwanta		Residential	0–30	3086	21,632.00
					30–300	4458	483,141.00
					300–600	256	98,043.00
					600+	33	29,675.00
					Total	**7833**	**632,491.00**
				Non Residential	0–30	742	3329.00
					30–300	453	44,389.00
					300–600	32	13,154.00
					600+	11	51,247.00
					Total	**1238**	**112,119.00**
				Low Voltage	0–30	0	0
					30–300	0	0
					Total	**0**	**0**
					30–300	0	0
				Medium Voltage	600+	0	0
					Total	**0**	**0**
					0–30	0	0
					30–300	0	0
				High Voltage			

Tables 1-5 and 1-4 illustrates the amount of electricity generation required to power the EVs. This includes accounting for electricity transmission and distribution losses, as well as battery charging inefficiency. The figures in this table serve as inputs for the subsequent calculations of well-to-wheel CO_2 emissions.

Table 1-6 provides insights into the average CO_2 emissions associated with electricity generation in the country. These figures help to understand the carbon intensity of the electricity grid, particularly the share of electricity generated from fossil fuels. The carbon intensity of electricity generation plays a significant role in determining the indirect well-to-wheels CO_2 emissions from EVs.

Table 1-7 provides an overview of the proportion of electricity generated from fossil fuels in Ghana. The table displays the specific figures for different energy sources, including oil, natural gas, hydro, renewables/other, and the net electricity generation for the country.

Table 1-8 presents the well-to-power-plant CO_2 emissions associated with power generation from different fuel sources. It demonstrates that the emissions from the electricity production phase, before reaching the power plant, are less than half the average emissions of coal, oil, and natural gas. Additionally, electricity generation from nuclear fission produces no CO_2 emissions. These figures contribute to the comprehensive evaluation of well-to-wheels CO_2 emissions for EVs.

Table 1-12
None-smart meter consumption for duayaw nkwanta (Source: NEDCo Duayaw Nkwanta).

Utility:		Duayaw Nkwanta	**Charging &YM:**	202,301	Samuel Sarquah		**Print Time:**	
AREA	**STATIONCODE**	**STATIONNAME**	**CONSUMPTION (kWh)**			**CUSTOMER NUMBERS**		
			RESIDENTIAL	**NORESIDENTIAL**	**TOTAL**	**RESIDENTIAL**	**NONRESIDENTIAL**	**TOTAL**
Sunyani	108	Duayaw Nkwanta	123,811.70	91,648.10	215,459.80	979	353	1332
	Total Consumption (kWh)		**123,811.70**	**91,648.10**	**215,459.80**	**979**	**353**	**1332**
Grand Total			**123,811.70**	**91,648.10**	**215,459.80**			

Table 1-13
EV population in Duayaw Nkwanta.

Daily Consumption (kWh/d/EV)	43.7421								
Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Population Duayaw Nkwanta	32,720.7	33,362.0	34,015.9	34,682.6	35,362.4	36,055.5	36,762.2	37,482.7	38,217.3
EV Growth rate/population	0.0009	0.0018	0.0026	0.0034	0.0041	0.0049	0.0056	0.0063	0.0069
Number of EVs in Duayaw Nkwanta	29	59	88.0	117	147	176	205	235	264
EV Consumption daily kWh/d	1282.7	2565.3	3848.0	5130.6	6413.3	7695.9	8978.6	10,261.3	11,543.9

Table 2-1
Electric vehicle population in 2030.

Year	2030
Conventional Car Population	4,040,000
EV Number in Ghana	270,000

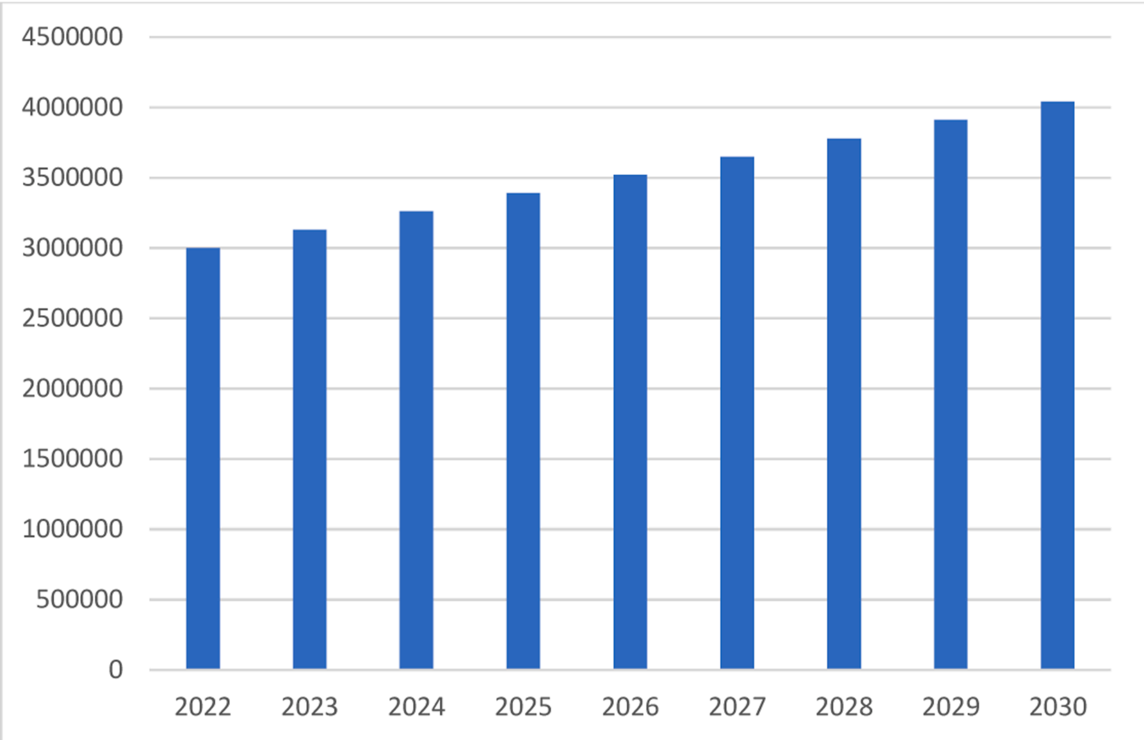

Fig. 2-1. Yearly vehicle population (Source: Authors' Results).

Impact of EV update on Duayaw Nkwanta distribution network

DIgSILENT PowerFactory has been used to analyse the grid at Duayaw Nkwanta to ascertain the impacts of the Electric Vehicle uptake. It looks at how the distribution lines (Cables) and transformer in the low-voltage network will be affected. This is very important for planning purposes as the government of Ghana has put in measures to increase the EV update. The goal of the analysis is to determine how well the current low-voltage grid can accommodate EV charging. Furthermore, to determine whether local grids need to be reinforced or whether any possible problems can be avoided by changing charging patterns and power.

Case study

To examine the impact of EV charging on the low-voltage grid, a case study was chosen. According to [48], general knowledge can be obtained from a single or a few cases if these cases are chosen using an information-based strategy. For this study, a Duayaw Nkwanta distribution area was chosen as a case study. The distribution network is shown in Fig. 1-1.

This method differs from the method in which examples are chosen at random and a statistically significant number of cases are

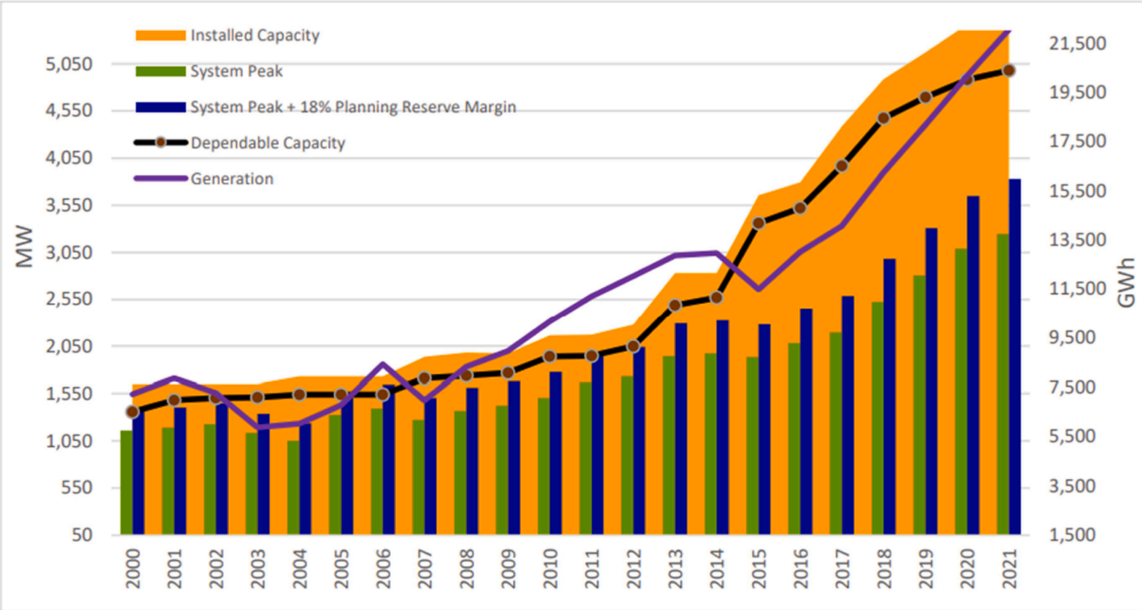

Fig. 2-2. Ghana's energy capacity [51].

Table 2-2
Demand Projection versus generation capacity (Source: Authors' Results).

Year	2023	2024	2025	2026	2027	2028	2029	2030
Yearly EV Consumption GWh	581	871	1162	1452	1743	2033	2323	2614
Projected Demand GWh	23,579	25,748	28,117	30,704	33,529	36,613	39,982	43,660
Total Demand (EV+Grid)	24,160	26,620	29,279	32,156	35,271	38,646	42,305	46,274
Generation (GWh)	42,711							

utilized to obtain knowledge that can be generalized [48]. Because power flow analysis is a time-consuming technique, only a few cases can be examined, hence the areas chosen for these cases must be information-based. The power flow study of Duayaw Nkwanta is selected to understand how stochastically the low voltage grid will behave. In terms of generalization, it should be highlighted that the attributes of electrical distribution networks are unique to the system under consideration. As a result, it can be misleading to relate the findings from two cases to all low-voltage distribution grids. However, the results of these scenarios can be used to show how low-voltage grids with similar characteristics perform. The following data was entered in the software as properties of the transformer and distribution lines as shown in Tables 1-9 and 1-10.

Electricity demand of Duayaw Nkwanta

The demand of Duayaw Nkwanta was obtained from the NEDCo Duayaw Nkwanta office to aid in determining the impact of the Electric Vehicle uptake on the local grid network. This consumption of the area is presented in Tables 1-11 and 1-12

The demand for Duayaw Nkwanta was calculated as shown in Table 1-13, this was calculated as shown in the table.

Demand profile generation

To simulate the impact of EV charging on a low-voltage grid, a demand profile of an EV charger and a household's electrical consumption must be generated. These profiles are used to simulate the electrical load of an average household and EV charger.

The household consumption profile will be fixed, and the EV charger demand will increase with EV share, as the aim is to assess the impact on the low voltage grid as EV power demand is added to the total demand.

Results and discussion

This section of the study presents the outcomes obtained and how they respond to the objectives of the study. It includes discussions of the results obtained. The objective of this research was to analyse the future growth and adoption of electric vehicles (EVs) in Ghana, assess the emission reduction potential of replacing conventional vehicles with electro-mobility, and evaluate the readiness of the existing electricity infrastructure in Ghana to support the projected increase in electro-mobility.

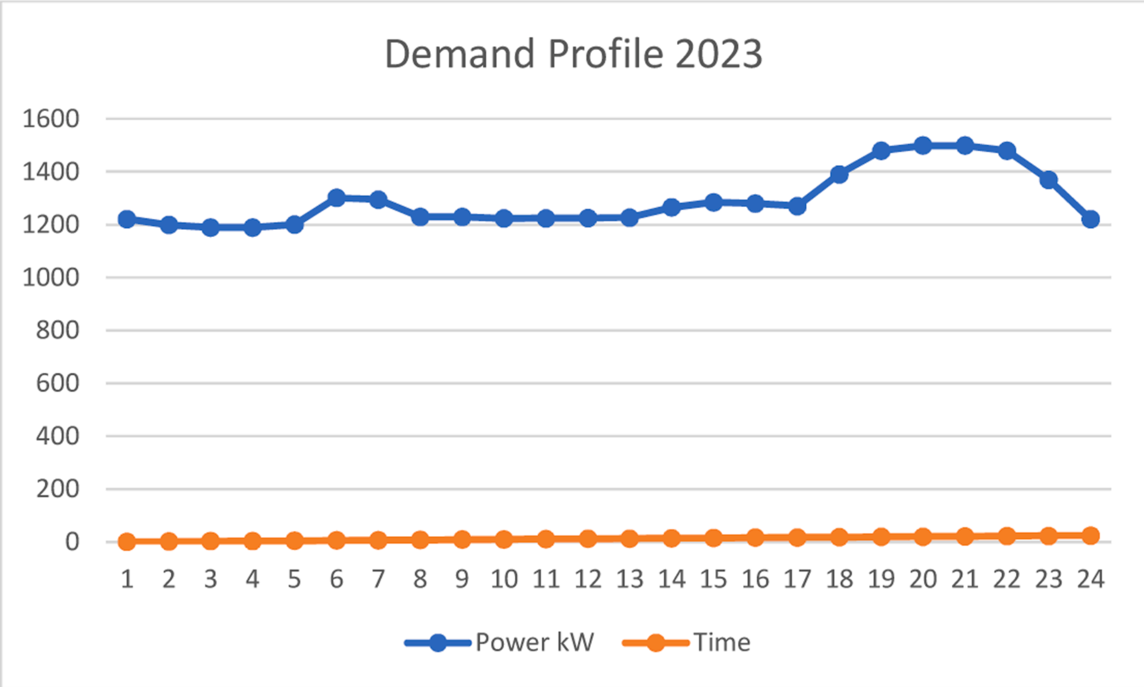

Fig. 2-3. Demand profile of Duayaw Nkwanta 2021 *(Source: Authors' Results)*.

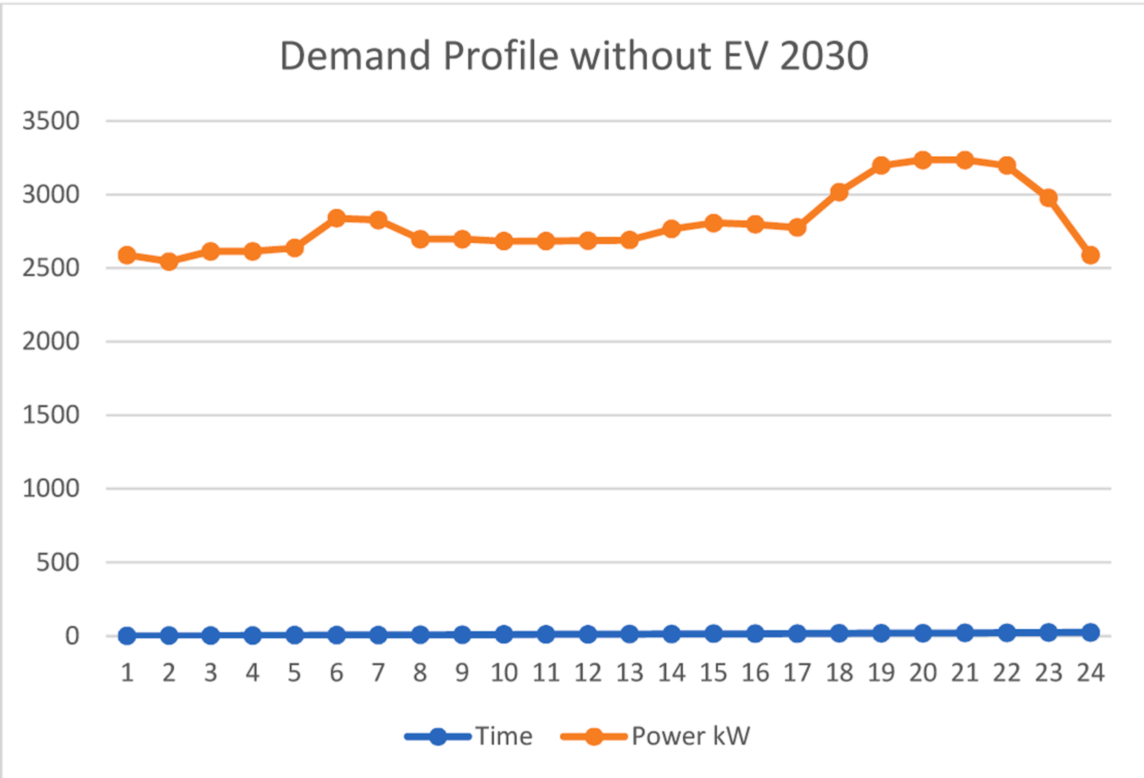

Fig. 2-4. Demand Profile without EV in 2030 (Source: Authors' Results).

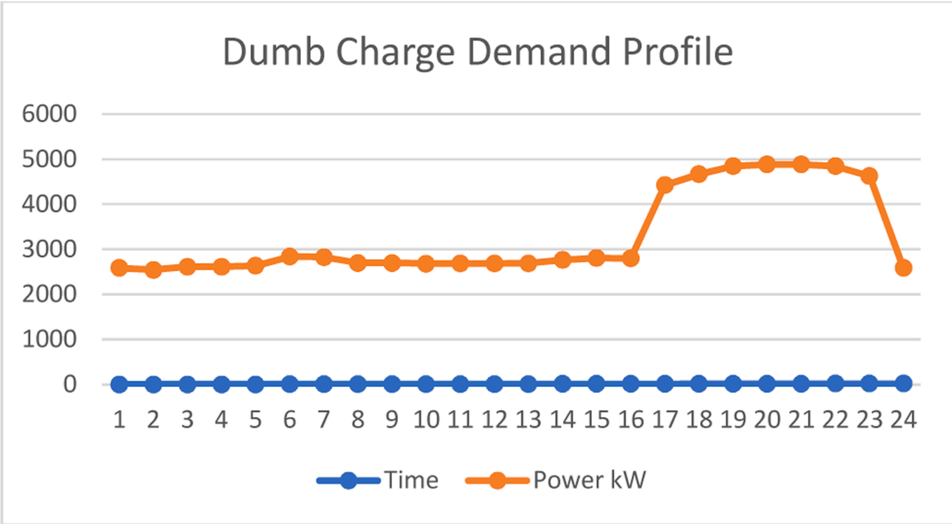

Fig. 2-5. Dumb charging Profile (Source: Authors' Results).

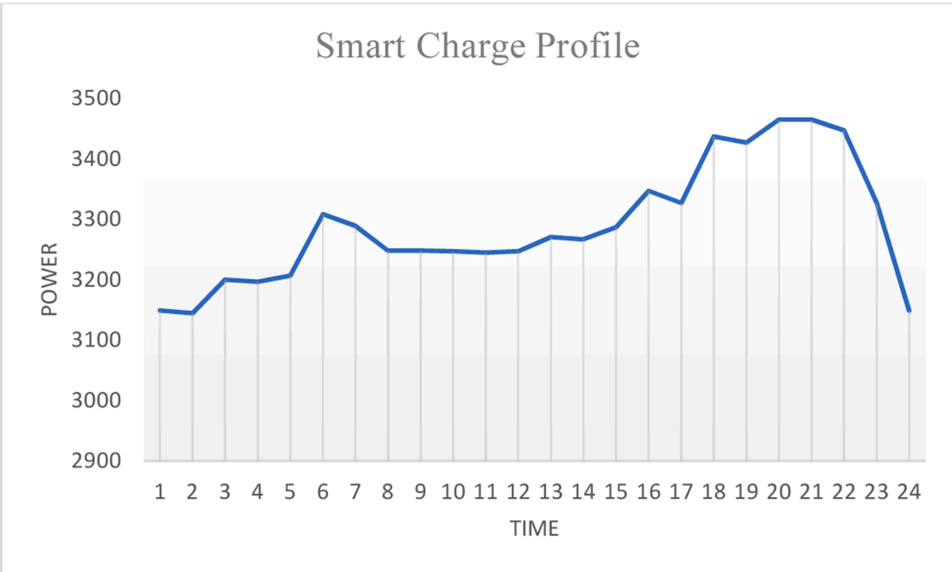

Fig. 2-6. Smart charging profile (Source: Authors' Results).

Electric mobility growth in Ghana

The study sought to estimate the number of electric vehicles that will be available in 2030 and estimate the emission reduction as well as the impacts on the national grid and the demand profile of Ghana. The results from the analysis as presented in Table 2-1 show that there will be about 270,000 EVs in Ghana representing 6.7 percent of the total vehicle population in Ghana. However, these figures could be changed depending on the incentives provided by the government.

Ghana experiences an annual growth in its vehicle population of approximately 30,000 as shown in Fig. 2-1, with the current vehicle count standing at approximately 3 million. This presents a substantial potential for the adoption of electric vehicles (EVs). Drawing from the Danish example, where the EV count grew from 219 in 2010 to 2919 in just five years, and surged to 15,507 in 2020, there is evident room for considerable EV uptake [49]. The replacement of 30,000 newly purchased vehicles each year with EVs has the potential to significantly increase adoption rates.

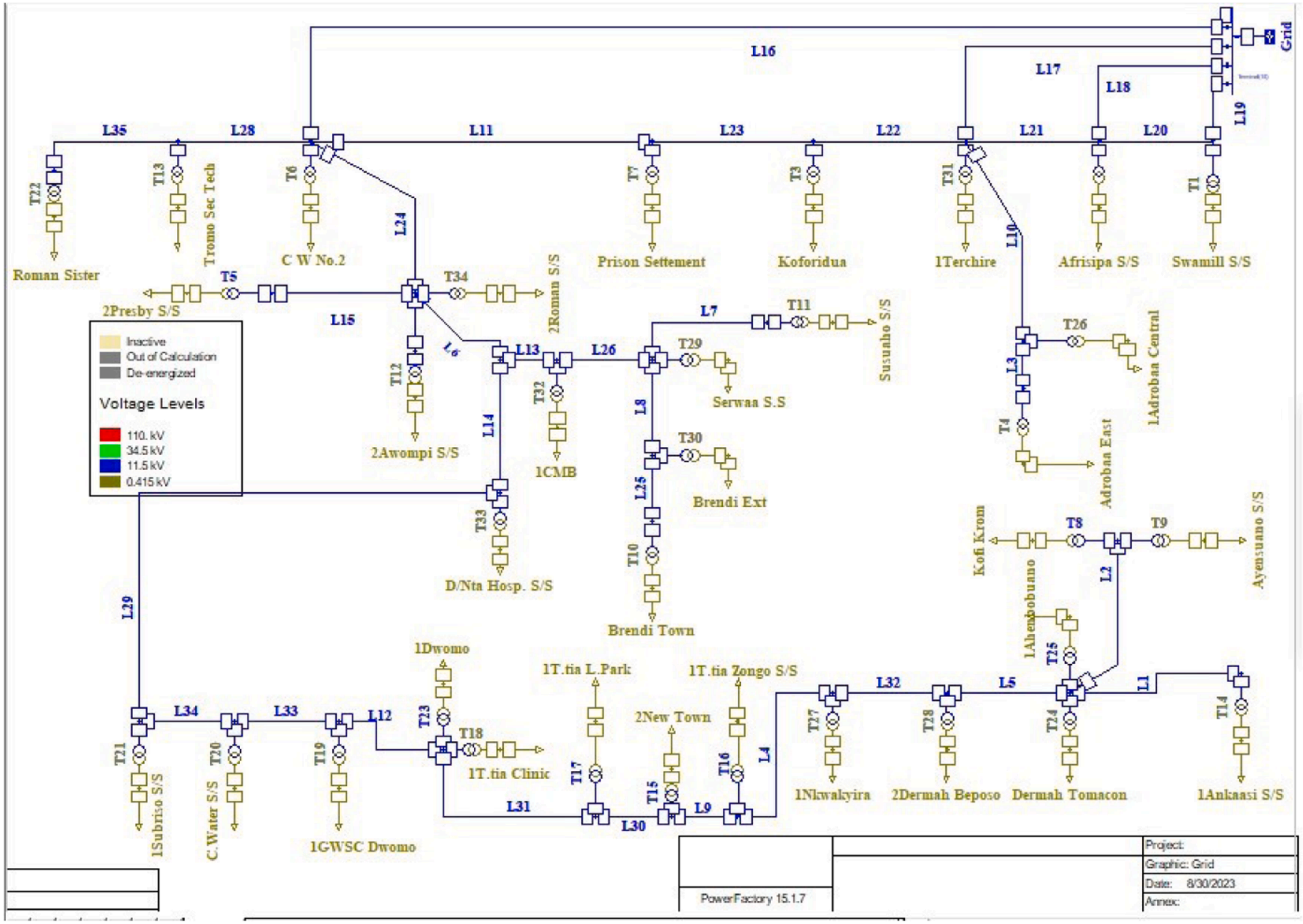

Fig. 2-7. Modelled Duayaw Nkwanta grid without simulation (Source: Authors' Results).

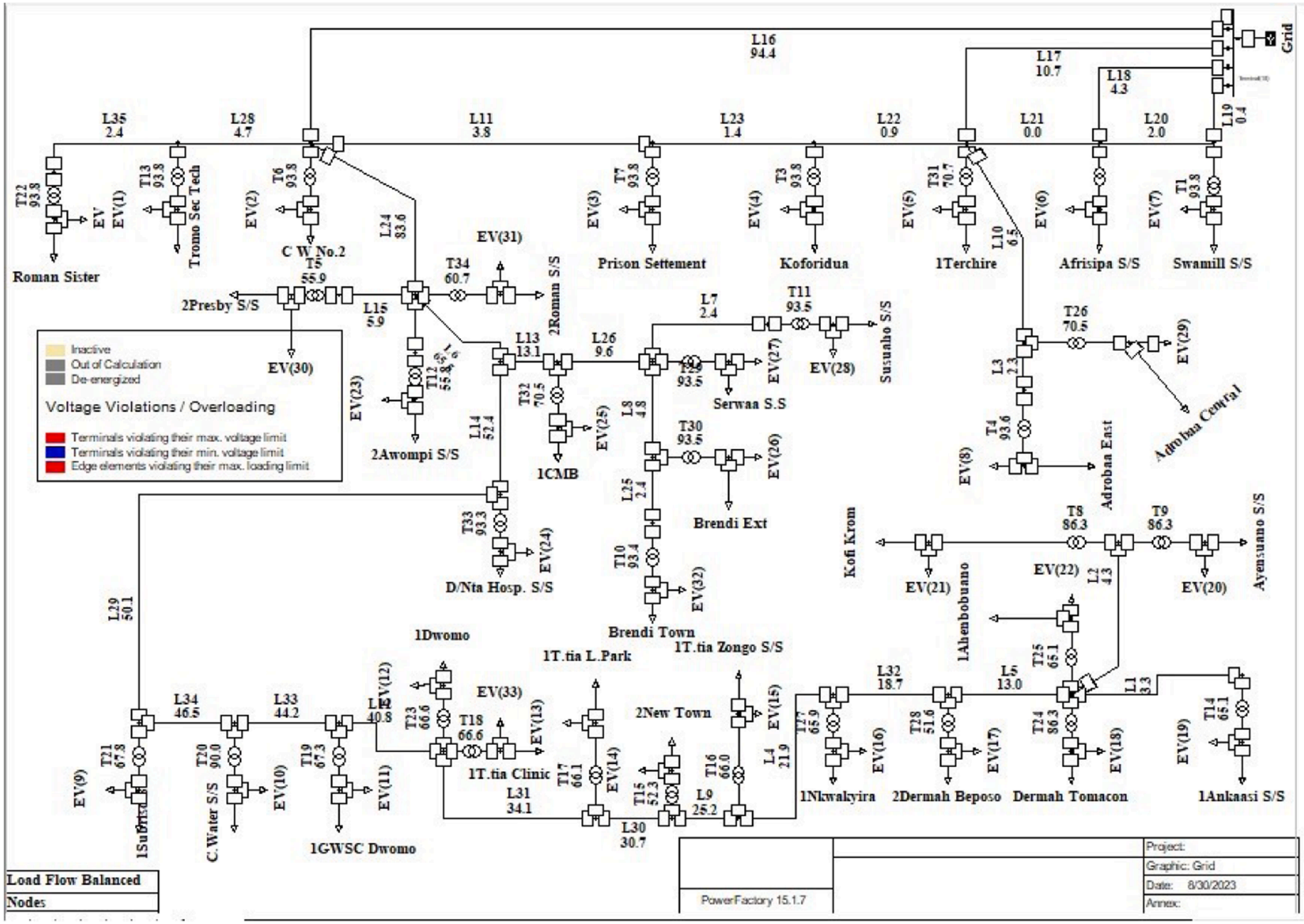

Fig. 2-8. Results of 2030 demand without EV (Source: Authors' Results).

Table 2-3
Load flow Results for 2030 demand without EV (Source: Authors' Results).

Parameter	Number	Parameter	Number
Substations	0	Shunts	0
2-w Trfs	34	Terminals	66
loads	34	Synchronous Machines	0
Busbars	1	SVS	0
3-w Trfs	0	Lines	35
		Asynchronous Machines	0

Parameter	Real power (MW)	Reactive Power (Mvar)	Apparent Power (MVA)
Generation	0	0	0
External Infeed	2.79	-0.47	2.83
Inter Grid Flow	0	0	
Load P(U)	2.63	0.01	2.63
Load P(Un)	2.85	0.01	2.85
Load P(Un-U)	0.21	0	
Motor Load	0	0	0
Grid Losses	0.01	-0.48	
Line charging		-0.69	
Compensation ind.		0	
Compensation Capacity		0	
Total power Factor:			
Generation	0.00 [-]		
Load/Motor	1.00/0.00[-]		

Demand and EV growth impact on electric demand in 2030

Ghana's electricity demand continues to rise, with an excess capacity gap expanding over time as shown in Fig. 2-2. The country's current installed generation capacity is 5481 MW, with a combination of hydro and thermal generation fuelled primarily by gas. As of 2021, the generating mix was around 34.65% renewables and 65.3% thermal [50]. Electricity generation nearly doubled from 11,200 GWh in 2011 to 22,051 GWh in 2021, a 7% yearly growth rate.

In 2021, the total power generated was 7643 GWh from renewable sources and 14,408 GWh from thermal generation. It is expected that generation will increase to 23,578.57 MWh, with renewable sources accounting for 7578.53 GWh and thermal sources accounting for 16,004 GWh. GRIDCo's supply plan for 2022. Peak demand in 2021 was 3246.0 MW, a 5% increase over the previous year. The 2021 high is a 156.0 MW increase above the 2020 peak of 3089.5 MW (a 5.0% increase). The country has enough generation to meet our short-term demands, with a predicted peak demand of 3545 MW and a projected dependable generation capacity of 4618 MW.

Table 2-2 shows the demand and supply outlook, it can be noted that the current installed capacity exceeds demand by about 20,000 GWh of electricity. However, with the rate of year-by-year increase in electricity demand, the current capacity will be able to meet the EV growth up to 2028 where plans must be put in place to augment the capacity. This means that the country has enough time to plan and put measures in place to continue sustaining the grid.

Grid impact analysis

The grid impact analysis conducted here represents a crucial dimension of the research, focusing on the implications of electric vehicle (EV) uptake on the electrical grid. Leveraging the capabilities of DIgSILENT Power Factory, a cutting-edge power system simulation tool, we have embarked on an assessment to measure the charging impact of EVs and their consequences on grid operations.

The grid impact analysis revolves around an intricate exploration of how the widespread adoption of electric vehicles affects the stability, capacity, and performance of the electrical grid. It encompasses an array of vital considerations, including load forecasting, charging behaviour modelling, grid capacity assessment, and the evaluation of potential bottlenecks or vulnerabilities that may arise as EVs proliferate.

Demand Profiles for the analysis

To study the grid, impact a case study was chosen as described. The demand for the location under study was critical and had to be determined. The demand was obtained from the grid owners manning the case study area (Duayaw Nkwanta). Fig. 2-3 shows the demand for the baseline year. The peak is between the hours of 7 pm and 10 pm and a much lower peak in the morning. This is because more energy-consuming appliances are turned on during these times. The demand profile is important because it gives an overview of when the grid is likely to experience a shortfall during EV charging.

Based on the electricity demand growth as discussed in Chapter Three, the demand for 2030 was forecasted or projected. Fig. 2-4 has a similar characteristic just that the magnitude in terms of power consumed has increased.

Fig. 2-5 depicts a dump charging profile. This means that people are inflexible in their charge times, they are assumed to also be charging at peak times. The purpose of the EV Dump Charge Scenario is to contemplate the effects of a situation where EVs are the new

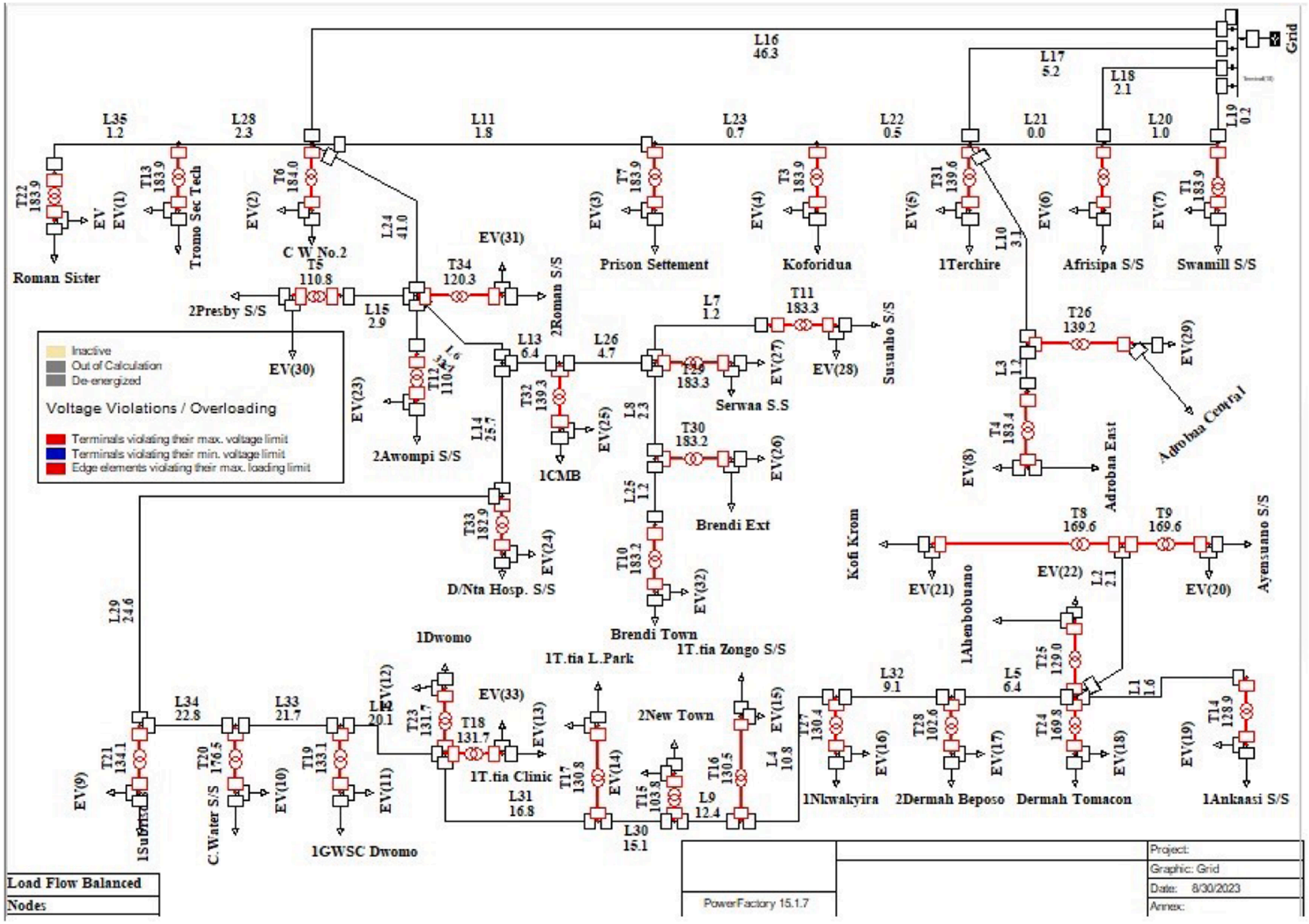

Fig. 2-9. Simulation results with EV included (Source: Authors' Results).

Table 2-4
Load flow results for 2030 demand without EV (Source: Authors' Results).

Parameter	Number	Parameter	Number
Substations	0	Shunts	0
2-w Trfs	34	Terminals	66
loads	68	Synchronous Machines	0
Busbars	1	SVS	0
3-w Trfs	0	Lines	35
		Asynchronous Machines	0

Parameter	Real power (MW)	Reactive Power (Mvar)	Apparent Power (MVA)
Generation	0	0	0
External Infeed	4.28	-0.16	4.28
Inter Grid Flow	0	0	
Load P(U)	3.92	0	3.92
Load P(Un)	4.49	0	4.49
Load P(Un-U)	0.57	0	
Motor Load	0	0	0
Grid Losses	0.36	-0.16	
Line Charging		-0.67	
Compensation ind.		0	
Compensation Capacity		0	
Total power Factor:			
Generation	0.00 [-]		
Load/Motor	1.00/0.00[-]		

normal and they add to the end-of-day electricity demand peak by recharging the vehicles after work.

Fig. 2-6 shows a smart charging profile. The purpose of the Smart Charge Scenario is to consider the consequences where flexible charging is considered an asset and 90% of charging needs are flexible.

Grid analysis results

This section of the study presents the results obtained from the DIgSILENT PowerFactory tool. Fig. 2-7 shows the grid network within the study area, Duayaw Nkwanta. It illustrates the positioning of all transformers and power lines, including their dimensional attributes. This visualization serves as a preliminary overview of the network configuration before initiating the simulation process. Notably, the transformers are designed to receive an input voltage of 34.5 KV, subsequently stepping it down to 415/220 V for distribution.

The network is then run with the 2030 load demand of the area to visualize the impacts on the grid and the results show that the grid can handle the demand of the area without any power issue. This system is without EV power demand. The active power was 2.85 MW. This is shown in Fig. 2-8 and Table 2-3: Load flow Results for 2030 demand without EV, the colors of the lines (black) indicate that the network is without any impacts that need attention.

The simulation with EVs included shows red indicators on the transformer while the power lines remain black as shown in Fig. 2-9. This means that the transformers are overloaded (turned to red) but the lines are okay. The active power has now increased to 4.49 MW as shown in Table 2-4. This is an indication that the grid will not be able to handle the penetration of EVs of that magnitude in 2030 as the local power demand is also increasing. Measures have to be taken to address the challenge if EV penetration is envisaged. The potential remedy is changing the transformer or having a parallel transformer to handle the loads as they increase. Even though the lines are not impacted, steps must be taken to increase their power-carrying capacities.

The system transformer underwent an augmentation by incorporating a dual transformer setup, allowing for parallel processing. This enhancement was simulated to evaluate its effects, as depicted in Fig. 2-10. The results of this simulation demonstrated the system's improved capacity to manage the heightened demand stemming from the increased adoption of electric vehicles (EVs). In the context of the electric vehicle increase, the integration of a parallel transformer architecture proved to be a pivotal upgrade. The dual transformers worked in tandem to distribute the load efficiently, ensuring that the system remained robust and responsive, even under the intensified demand caused by the growing number of EVs on the network. This optimization highlights the adaptability of the system to evolving technological and environmental challenges, displaying its readiness to meet the demands of a changing energy landscape. However, this approach can be very expensive. Another compelling option is to introduce smart charging approach so that EV owners can charge their vehicles during off-peak hours. This approach cab be achieved by incentivizing them to reduced tariff.

Emission analysis results

The widespread adoption of Electric Vehicles (EVs) represents a transformative shift in the transportation sector, promising substantial reductions in greenhouse gas emissions and a path toward a more sustainable and environmentally responsible future. As governments and industries around the world invest and promote electric vehicles it becomes imperative to assess the net environmental impact of integrating EVs into the existing energy infrastructure.

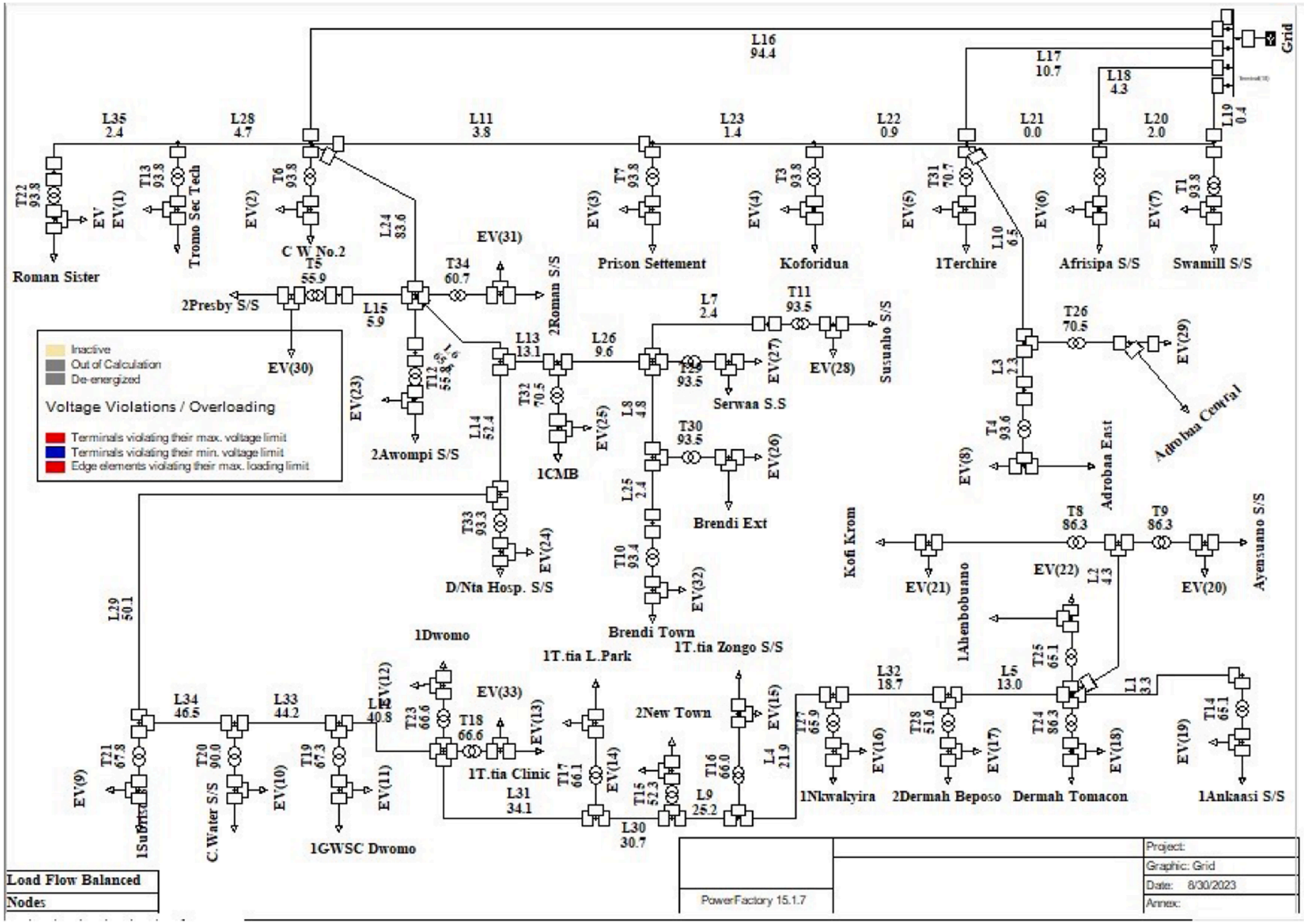

Fig. 2-10. Transformer problem solved (Source: Authors' Results).

Table 2-5
Emission comparison (Source: Authors' Results).

Vehicle Type	Emission Levels gCO2/km
Evs	62.803
PHEVs	116.789
Petrol/Diesel	198.803

To gain a holistic understanding of the environmental implications associated with EV adoption, a "Well-to-Wheel" (WTW) approach was used as discussed.

These emission figures in Table 2-5 illustrate a clear advantage of Electric Vehicles (EVs) over both Plug-in Hybrid Electric Vehicles (PHEVs) and traditional Petrol/Diesel vehicles in terms of carbon dioxide emissions per kilometer traveled. EVs emit only 62.803 gCO_2/km, signifying a significant reduction in greenhouse gas emissions compared to their counterparts This is even because of the source charging in Ghana which is dominated by fossil-fueled generators. PHEVs, while offering some emissions reduction benefits due to their electric mode, still produce nearly double the emissions of EVs at 116.789 gCO_2/km. Meanwhile, conventional Petrol/Diesel vehicles emit a strikingly higher 198.803 gCO_2/km, emphasizing the urgency of transitioning to cleaner transportation alternatives like EVs to combat climate change and reduce the overall carbon footprint of our transportation systems. These figures underscore the importance of promoting EV adoption as a key strategy for reducing emissions.

Considering the study area Duayaw Nkwanta which is projected to have 264 EVs in the year 2030, an amount of 35,904 gCO2/km will be saved.

Conclusions

The study aimed to project the landscape of electric vehicle (EV) adoption in Ghana by 2030 and assess its associated impacts on emissions, the national grid, and the country's energy demand profile. The analysis conducted indicates that by 2030, Ghana could potentially have approximately 270,000 EVs on its roads, representing a significant shift toward cleaner transportation, amounting to 6.7% of the total vehicle population.

However, it is crucial to acknowledge that these projections are subject to modification based on government incentives and policies designed to encourage EV adoption. Drawing inspiration from the rapid growth of EVs in Denmark, we recognize a substantial opportunity for Ghana to increase its EV uptake, particularly by replacing a portion of the 30,000 newly purchased vehicles annually with EVs.

The study also delved into the dynamics of Ghana's energy landscape, showcasing a steady increase in electricity demand, and highlighting an excess capacity gap. Although the current electricity generation infrastructure seems sufficient to meet short-term needs, it is essential to plan, especially as EV adoption gains momentum. The simulation results emphasize the need for augmenting the grid's capacity, including introducing parallel transformers and reinforcing power lines, to accommodate the anticipated EV growth.

While Ghana has the potential to embrace electric vehicles as a sustainable mobility solution, careful planning and infrastructure development are essential to ensure that the nation can effectively integrate EVs into its transportation system while maintaining a stable and reliable electrical grid. This research provides valuable insights for policymakers, utility companies, and stakeholders to make informed decisions and take proactive measures to support the sustainable growth of EVs in Ghana.

The shift toward electric mobility is a gradual process, not an overnight transformation, and the widespread adoption of electric vehicles (EVs) will be staggered. Consequently, the existing charging infrastructure's current capacity should be sufficient to meet demand requirements. However, the crucial aspect lies in the timing of charging events. Unregulated charging behaviour has the potential to strain peak demand and impose challenges on the grid. Hence, it is imperative to implement targeted policies that encourage the reduction of uncontrolled or 'dumb' charging while promoting intelligent charging practices. This entails incentivizing off-peak charging, such as during night-time hours or after 10 p.m., and introducing pricing mechanisms that discourage peak-time charging. Moreover, raising awareness about responsible charging practices is essential to ensure the harmonious integration of EVs into our transportation ecosystem.

Again, the grid must have a higher percentage been renewable energy other than that it would be the same as using petrol or diesel cars.

One of the primary inconveniences associated with Electric Vehicles (EVs) is the extended charging time required, discouraging many potential users from embracing this technology. Additionally, the shortage of charging infrastructure and its proximity to residential areas compounds this issue. Given these challenges, we strongly advocate for the expansion of charging station networks. To enhance user convenience, it is imperative to install more charging stations equipped with efficient fast-charging capabilities at strategically advantageous locations, ensuring that EV owners can readily access them.

CRediT authorship contribution statement

Emmanuel Yeboah Asuamah: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing. **Abass Ishaq:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Samuel Jonas Yeboah:** Investigation, Writing – review &

editing, Supervision, Project administration. **William Duodu Asihene:** Investigation, Writing – review & editing, Project administration.

Declaration of competing interest

We declare that the work described has not been published previously, is not under consideration for publication elsewhere, is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out, and that, if accepted, will not be published elsewhere in the same form, in English or any other language, including electronically without the written consent of the copyright holder. We confirm that no known conflicts of interest are associated with this publication. The manuscript has been read and approved by all named authors.

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